

IMO 2026 — Problem 1*Solution by Meta Muse Spark 1.1*

Source: results/muse-spark-1.1/problem-01/current.md. This model's markdown contained unfenced/malformed LaTeX; the solution is shown as verbatim source below (the current.md file is the authoritative version).

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# imo-2026-01 – tracking file
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## Status
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solved
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## Problem
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There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

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## Approaches tried
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- Translate operation into prime exponent map $(a, b) \mapsto (\min(a, b), |a - b|)$, i.e. Euclidean subtraction.
- Observe total product $P = \prod x_i$ satisfies $P_{\text{new}} = P_{\text{old}} / \gcd(m, n)$.
- Observe $k_t = \#\{i: x_i > 1\}$ is nonincreasing, drops by 1 iff $\gcd = 1$ or $m = n$, otherwise stays.
- Construct infinite-descent contradiction: if k stabilizes at ≥ 2 , P strictly decreases forever.
- Per prime invariant $G_p = \gcd_i v_p(x_i)$ unchanged by operation, yielding final M formula.

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## Current best
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For each prime p write v_p . Then $(m, n) \mapsto (\gcd, \operatorname{lcm} / \gcd)$ acts as $(a_p, b_p) \mapsto (\min, |a - b|)$. Hence total product P divides by $\gcd(m, n)$ each move and $k_t = \#\{x_i > 1\}$ never increases; if a sequence were infinite k_t would stabilize at ≥ 2 after which P would strictly decrease infinitely, impossible. Thus process always stops, and it can stop only when $k = 1$, giving a single $M > 1$. Moreover $G_p = \gcd_i v_p(x_i)$ is Euclidean-invariant, so in final state $v_p(M) = G_p$, i.e. $M = \prod p^{\gcd_i v_p(x_i^{(0)})}$, independent of choices.

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## Full proof
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Let $N = 2026$; argument works for any $N \geq 2$. Denote the multiset after t moves by

$x^{(t)}_1, \dots, x^{(t)}_N, x^{(0)}_{i > 1}$.

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### 1. The operation in prime exponents.
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For $m, n > 1$ write $m = \prod p^{a_p}$, $n = \prod p^{b_p}$ ($a_p, b_p \geq 0$). Then

$$v_p(\gcd(m, n)) = \min(a_p, b_p), \quad v_p(\operatorname{lcm}(m, n)) = \max(a_p, b_p).$$

Hence

$$\begin{equation} \tag{1} \end{equation}$$

$$v_p\left(\frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}\right) = \max(a_p, b_p) - \min(a_p, b_p) = |a_p - b_p|.$$

$$\end{equation}$$

Also $\operatorname{lcm}(m, n) = mn / \gcd(m, n)$, so

$$\begin{equation} \tag{2} \end{equation}$$

$$\gcd(m, n) \cdot \frac{\operatorname{lcm}(m, n)}{\gcd(m, n)} = \operatorname{lcm}(m, n) = \frac{mn}{\gcd(m, n)}.$$

$$\end{equation}$$

Thus on exponents pair (a_p, b_p) the move replaces (up to order) by

$$\begin{equation} \tag{3} \end{equation}$$

$$(\min(a_p, b_p), |a_p - b_p|),$$

$$\end{equation}$$

the subtractive Euclidean step.

Lemma: not both replacements are $\$1$. Indeed if $g = \gcd(m, n) = 1$ then

$\ell/g = mn > 1$. If $\ell/g = 1$ then $\ell = g$; but $m \nmid \ell$ and $n \nmid \ell$,

so $m \nmid \ell$ and $n \nmid \ell$, while $g \mid m, n$; thus $m = n = g > 1$. So with

$m \neq n$ impossible to have $\ell/g = 1$ when $g > 1$; check: $\ell/g = 1 \Rightarrow \ell = g \Rightarrow m = n = g$ as above. Consequently:

- If $g = 1$, new pair is $\{1, mn\}$, exactly one > 1 .
- If $g > 1$ and $m = n$, then $\ell = m = n = g$, $\ell/g = 1$, new pair $\{g, 1\}$, exactly one > 1 .
- If $g > 1$ and $m \neq n$, then both $g > 1$ and $\ell/g > 1$, new pair has two > 1 .

In particular at least one new number is > 1 .

2. Monotone quantities.

Product: $P_t = \prod_{i=1}^N x^{(t)}_i \in \mathbb{N}$. By (2), if move t to $t+1$ uses m, n with $g = \gcd(m, n)$,

$$\begin{equation} \tag{4} \end{equation}$$

$$P_{t+1} = P_t \cdot \frac{g \cdot \ell}{mn} = P_t \cdot \frac{\ell}{mn} = P_t / g.$$

$$\end{equation}$$

So $P_{t+1} \leq P_t$, with equality iff $g = 1$, strict iff $g > 1$.

Count: $k_t = |\{i: x^{(t)}_i > 1\}|$. From classification above, for a move on two > 1 entries:

$$k_{t+1} =$$

$$\begin{cases} \end{cases}$$

$$k_t - 1, \text{ \& } g = 1,$$

$$k_t - 1, \text{ \& } g > 1, m = n,$$

$$k_t, \text{ \& } g > 1, m \neq n.$$

$$\end{cases}$$

Indeed we remove two > 1 and insert $\{1, 2\}$ numbers > 1 .

Thus $k_{t+1} \leq k_t$ and $k_{t+1} \geq k_t - 1$.

Since initial $k_0 = N \geq 2$ and each move keeps at least one > 1 among the replaced pair,

k_t can never become 0 : if $k_t \geq 2$, $k_{t+1} = k_t - 2 + r \geq k_t - 1 \geq 1$.

If $k_t = 1$ there is no legal move. Hence throughout defined process $k_t \geq 1$,

nonincreasing.

3. Finiteness and form of terminal state – (a).

Assume an infinite sequence of moves existed. k_t nonincreasing integer bounded below by 1 , so eventually constant: $\exists T$ with $k_t = k$ for all $t \geq T$.

If $k = 1$, no two > 1 entries exist, process would have stopped at T , contrary to infinitude.

Thus $k \geq 2$ for $t \geq T$.

For $t \geq T$, $k_{t+1} = k_t$, so every move after T is of the third type:

$m \neq n$, $g > 1$, both replacements > 1 . Then by (4) $P_{t+1} = P_t / g < P_t$.

Thus $(P_t)_{t \geq T}$ is a strictly decreasing infinite sequence of positive integers, impossible.

Hence no infinite sequence exists; every choice sequence terminates after finitely many moves.

A terminal state means fewer than two entries > 1 . Since $k_t \geq 1$ always, termination implies $k_t = 1$. So exactly one integer > 1 remains, the other $N - 1$ entries are 1 . This proves (a).

4. Invariant – (b).

Fix prime p . Let $a^{(t)}_i = v_p(x^{(t)}_i) \geq 0$. Define

$$[G^{(t)}]_p = \gcd(a^{(t)}_1, \dots, a^{(t)}_N)$$

with $\gcd(0, \dots, 0) = 0$.

Claim: $G^{(t)}_p$ is invariant.

Indeed (3) replaces a_i, a_j by $a'_i = \min(a_i, a_j)$, $a'_j = |a_i - a_j|$.

For any $a, b \geq 0$,

$$\gcd(a, b) = \gcd(\min(a, b), |a - b|),$$

since if $a \leq b$, $\gcd(a, b) = \gcd(a, b - a)$ ($= \gcd(a, 0) = a$ if $a = b$ or $b = 0$), symmetrically.

Let R be the other $N - 2$ numbers. Then

$$\gcd(\text{all before}) = \gcd(\gcd(a_i, a_j), R) = \gcd(\gcd(a'_i, a'_j), R) = \gcd(\text{all after}).$$

Thus $G^{(t+1)}_p = G^{(t)}_p$.

Let $G_p = G^{(0)}_p$ be initial.

Consider final state after termination: $y_k = M + 1$ at some position k , $y_i = 1$ for $i \neq k$.

Then $a^{(\text{final})}_i = 0$ for $i \neq k$ and $a^{(\text{final})}_k = v_p(M)$. Hence

$$G^{(\text{final})}_p = \gcd(0, \dots, 0, v_p(M), 0, \dots, 0) = v_p(M)$$

(with $v_p(M) = 0$ allowed, giving 0). By invariance,

$$\begin{equation} \tag{5} \end{equation}$$

$$v_p(M) = G_p = \gcd_{1 \leq i \leq N} v_p(x^{(0)}_i) \quad \text{for every prime } p.$$

$\end{equation}$

Hence

$$M = \prod_p p^{\gcd_i v_p(x^{(0)}_i)},$$

where product over primes dividing some initial number (finite). This depends only on initial multiset, not on choices.

In particular existence of such $M > 1$: if $p \mid x^{(0)}_{i_0}$ with $v_p > 0$, then $\gcd_i v_p$ equals $\gcd$ of a collection containing 0 's and a positive number, so it is $\gcd$ of the positive part, ≥ 1 ; indeed $\gcd(a, 0) = a$. So any prime occurring in initial data contributes exponent ≥ 1 ; since at least one initial number > 1 has a prime factor, the product above is > 1 , consistent with (a). The final $M > 1$ is thus uniquely determined.

Therefore the value M reached when the process stops does not depend on the sequence of choices of Confucius. This proves (b). $\square$